

PGP Tools (Offline) — Technical Documentation & Guides

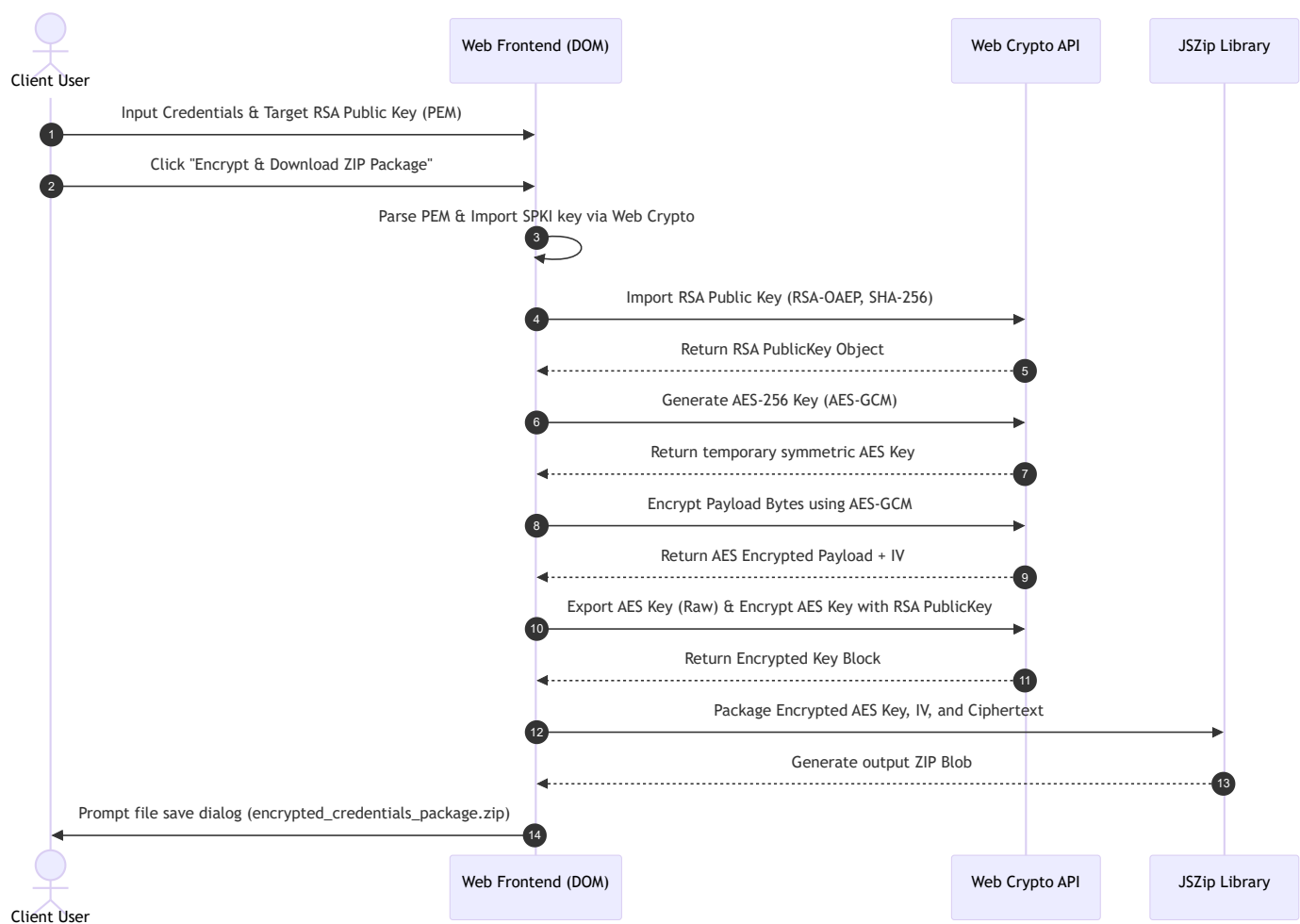
This repository contains the **API PGP Tools**, a security-focused, fully offline single-page web utility that packages, encrypts, and decrypts sensitive API credentials. The application utilizes client-side Web Cryptography APIs and OpenPGP.js to guarantee zero-trust data security.

Core Features & Functionality

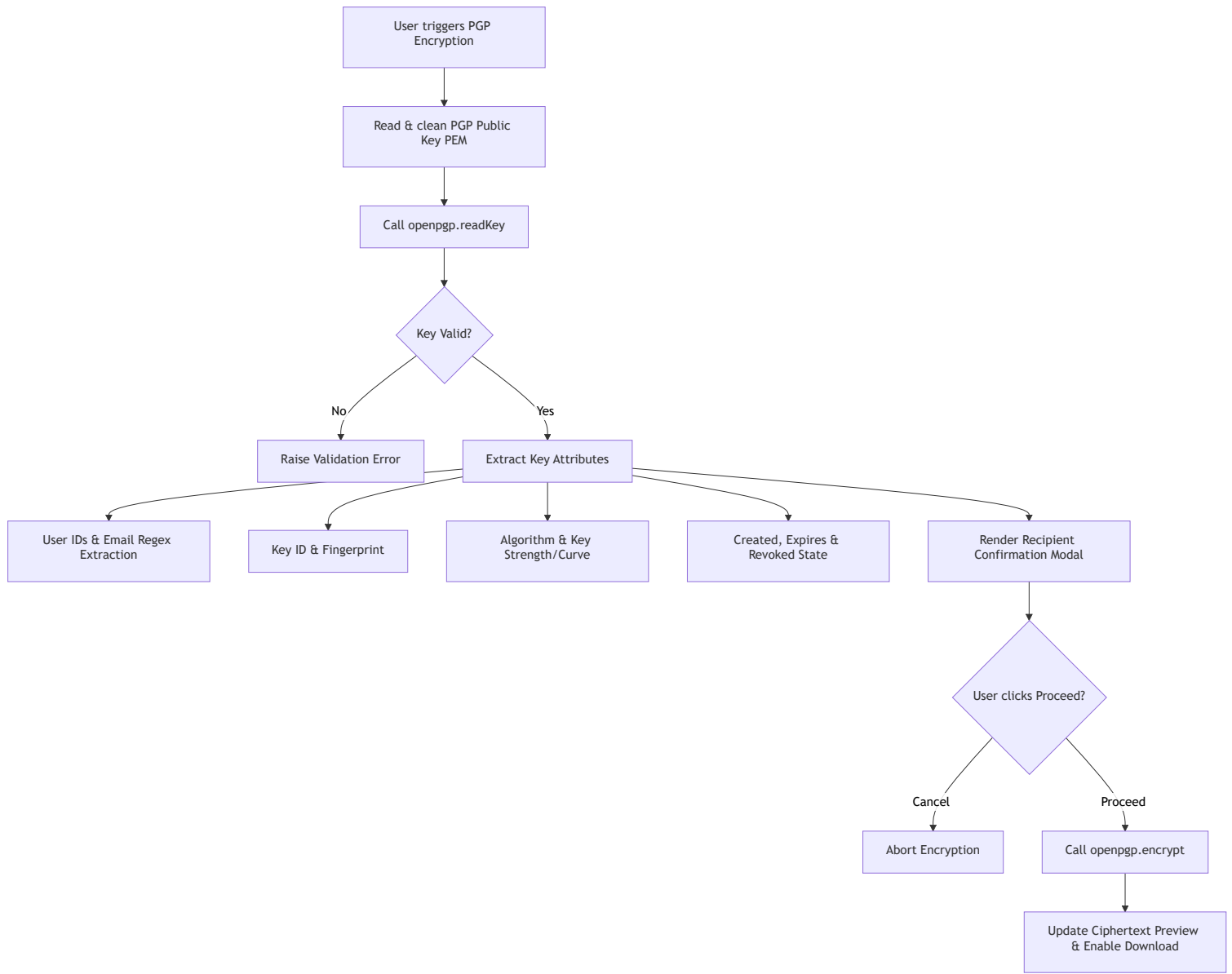
1. **Zero Network Connectivity:** Content-Security-Policy (CSP) restricted to prevent external network transactions.
2. **Hybrid RSA & AES Cryptography:** Packages plaintext JSON or generated credentials via `RSA-OAEP` (SHA-256) combined with `AES-GCM` (256-bit key) inside a secure ZIP.
3. **OpenPGP Encryption & Decryption:** Armored PGP Message encrypt/decrypt utilizing local ECC/RSA keys.
4. **Interactive Recipient Details Panel:** A confirmation modal parsed in real-time that validates key properties (User ID, Emails, Strength, Expiry, Revocation, Fingerprint) before performing any encryption.
5. **Secure In-Memory Credentials Auto-Generation:** Generates partner-id, secret, and client-id in memory as cryptographically strong UUIDs and immediately encrypts them without exposing plain text in the DOM or logs.
6. **Passphrase Dialog Verification:** Prompts user for private key password inside a masked modal, validating inline using OpenPGP decryption keys without closing the window or reloading state on failure.
7. **Configurable Security Session Timeout:** Auto-wipes browser memory and logs after a customizable inactivity interval.

Technical Architecture & Operations

1. Hybrid RSA-OAEP + AES-GCM Encryption Sequence



2. PGP Encryption & Metadata Extraction Flow



Key Configurations

Local Dependency Requirements

For strict offline verification, the following libraries must be present in the same directory:

- `openpgp.min.js` : Parses keys, manages passphrases, and performs PGP operations.
- `jszip.min.js` : Handles file packaging for RSA Hybrid archives.

Their validation and version information are displayed dynamically in a table at the top of the interface:

- **Status:** Loaded / Missing
- **Library:** Package Name
- **Version:** Version parsed from library properties
- **Filename:** Matching target script name